

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

ELEMENTARY CONCEPTS OF ELECTRICAL SYSTEMS

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence. Each question carries 1 mark

II. (9 x 1 = 9 Marks)

1	Define electric shock.	M1.01	R
2	Name any two electrical equipment that comes under the mandatory star labelling program.	M1.04	R
3	List any two modes of heat transfer.	M2.02	R
4	List any two industrial applications of electric heating.	M2.03	R
5	State Faraday's first law of electrolysis.	M2.04	R
6	Give any two examples of storage batteries.	M3.01	R
7	Materials whose resistance drops abruptly to zero at a temperature below its critical temperature are called -----.	M3.04	R
8	The property of magnetic materials which allows the flux to pass through it is called -----.	M4.01	R
9	Define capacitance of a capacitor.	M4.04	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Explain the phenomenon of electric shock	M1.01	U
2	List any six safety precautions to be followed while handling electricity.	M1.02	R
3	Illustrate the schematic diagram of an RCCB.	M1.02	U

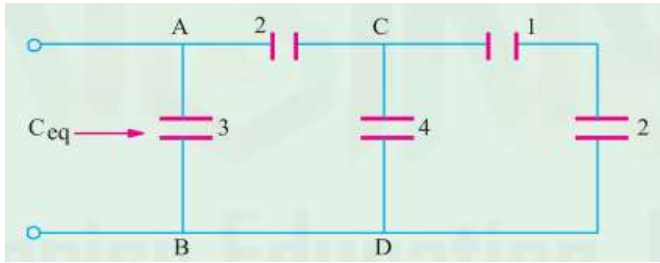
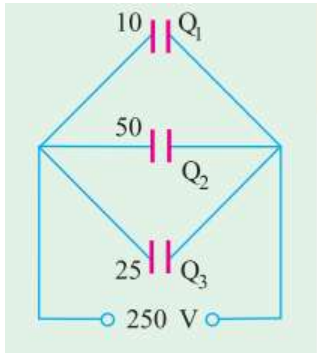
4	An electric iron takes 4 amperes at 250 volts. Determine the power rating of this appliance and find the number of units consumed if the iron is used for 90 minutes.	M1.03	A
5	List any six requirements of a good heating element.	M2.02	R
6	Illustrate the construction of a lead acid battery.	M3.02	U
7	Identify any three indications of a fully charged lead acid battery.	M3.02	U
8	Enumerate any six factors affecting capacity of battery.	M3.02	R
9	Select a suitable material for electrical wires and list any five electrical properties.	M3.04	R
10	State coulomb's laws of magnetic force and write its mathematical expression.	M4.01	R

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Prepare the detailed specification table of following electrical equipment; (i) Ceiling fan, (ii) Washing machine. OR	M1.03	A
IV	Illustrate BEE star labeling and identify various descriptions.	M1.04	U
V	Illustrate the construction and working of a fluorescent lamp. OR	M2.01	U
VI	Explain with a neat sketch the working principle of an LED lamp.	M2.01	U
VII	Outline the basic principle of electrolysis with a neat sketch. OR	M2.04	U
VIII	Explain the principle of dielectric heating with diagrams.	M2.03	U
IX	Summarize any seven caring measures to be adopted for maintenance of lead acid batteries. OR	M3.02	R
X	Summarize the classification of insulators based on its physical state with examples.	M3.04	U

XI	Find the equivalent capacitance of the circuit shown in fig. All capacitances are in μF. Show all steps.  OR	M4.04	A
XII	Three capacitors A, B, C have capacitances 10, 50 and 25 μF respectively. Calculate (i) charge on each when connected in parallel to a 250V supply (ii) total capacitance. 	M4.04	A
XIII	Explain the B-H curve with a neat sketch. OR	M4.01	U
XIV	Derive the expression for the equivalent capacitance when a group of capacitors are connected in series.	M4.04	U

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
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ELEMENTARY CONCEPTS OF ELECTRICAL SYSTEMS

MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	List any two effects of electric shock.	M1.01	R
2	Name the nodal agency responsible to implement the star labelling program in India.	M1.04	R
3	Enumerate the types of a sodium vapour lamp.	M2.01	R
4	Define convection mode of heat transfer.	M2.02	R
5	The process of depositing a coating of one metal over another metal or nonmetal electrically is called -----.	M2.04	R
6	Define ampere hour efficiency of a battery.	M3.01	R
7	Match the following (i) Copper - (a) Insulator (ii) Bakelite - (b) Ferromagnetic material (iii) Cobalt - (c) Paramagnetic material (iv) Magnesium - (d) Conductor	M3.04	R
8	State Faraday's first law of electromagnetic induction.	M4.02	R
9	The region in which the electric line of force acts around a charge is called -----.	M4.03	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	List the emergency procedures to be followed to rescue a person injured by an electric shock.	M1.01	R
2	Explain the operation of fuse in an electric circuit.	M1.02	U

3	Prepare the detailed specification table of an induction cooktop.	M1.03	A
4	Explain any six energy saving tips for a refrigerator.	M1.04	U
5	Draw the schematic diagram of sodium vapour lamp and label the parts.	M2.01	U
6	List any six advantages of electric heating.	M2.02	R
7	Illustrate direct resistance heating method.	M2.03	U
8	Specify the factors affecting the quality of electro deposition process.	M2.04	R
9	Enumerate any six applications of lead acid battery.	M3.02	R
10	State and explain the Coulomb's laws of electrostatics.	M4.03	R

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Following is the electricity consumption for an office: a. 10 fluorescent lamps each 40 watts working 8 hours/day. b. 4 fans each 60 watts working 8 hours/day. c. 2 coolers each 100 watts working 4 hours/day. d. 5 lamps each 40 watts working 2 hours/day. Calculate the electricity bill for the month of April at the rate of Rs.5 per unit. OR	M1.03	A
IV	Explain the trip mechanism in MCCB under the following fault conditions: (a) Overload, (b) Short circuit.	M1.02	U
V	Define electrolysis and summarize any four applications of electrolysis. OR	M2.04	R
VI	Illustrate the working of an incandescent lamp.	M2.01	U
VII	Classify insulators based on temperature and mention the maximum operating temperature. OR	M3.04	U

VIII	Explain with necessary sketches the operation of storage batteries with series and parallel connected cells.	M3.03	U
IX	Explain the various charging systems of batteries.	M3.02	U
	OR		
X	Classify and explain magnetic materials.	M3.04	U
XI	Two capacitors 8 μ F and 4 μ F are connected in parallel and a third of 24 μ F is connected in series with them. The combination is connected to 80V DC. Draw the arrangement of capacitors and find equivalent capacitance and charge.	M4.04	A
	OR		
XII	A set of three capacitors having capacitances of 10, 5 and 1 microfarad respectively is available. What is the value of the greatest and the smallest capacitance that can be obtained from the set and draw the arrangements.	M4.04	A
XIII	Compare electric and magnetic circuits.	M4.02	U
	OR		
XIV	With necessary sketches derive the equation for equivalent capacitance when a number of capacitors are connected in parallel.	M4.04	U

Scoring Indicators

Model Question Paper I

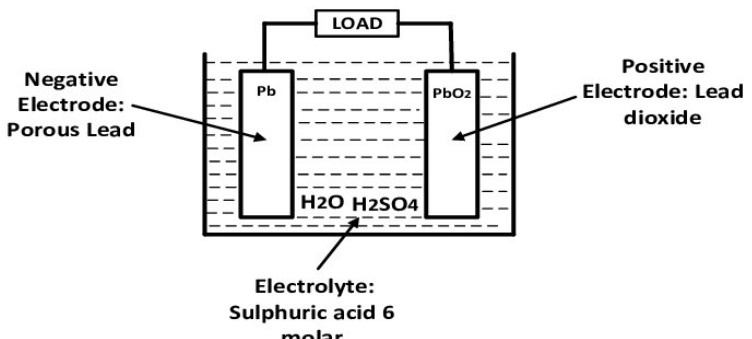
ELEMENTARY CONCEPTS OF ELECTRICAL SYSTEMS

Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			9
I. 1	<u>Electric shock</u> <i>Definition- 1 mark</i> An electrical shock is a physical reaction to electrical currents passing through the body.	1	1	
I. 2	<u>Equipment with mandatory star labelling program</u> <i>list-any two 2* 0.5 marks = 1 mark</i>	0.5+0.5	1	
	● Household Frost Free Refrigerators ● Room Air Conditioners ● Tubular Fluorescent Lamps ● Distribution Transformers (up to 200 KVA)			
I. 3	<u>Modes of heat transfer</u> <i>list-any two 2* 0.5 marks = 1 mark</i>	0.5+0.5	1	
	● Conduction ● Convection ● Radiation			
I. 4	<u>Industrial applications of electric heating</u> <i>list-any two 2* 0.5 marks = 1 mark</i>			
	● Melting of metals ● Molding of glass ● Baking of insulators ● Welding ● Enameling of copper wires	0.5+0.5	1	

I. 5	<u>Faraday's first law of electrolysis</u> <i>Statement - 1 mark</i>	1	1	
	The mass of a substance liberated from an electrolyte in a given time is proportional to the quantity of electricity passing through the electrolyte.			
I. 6	<u>Examples of storage batteries</u> <i>list-any two</i> <i>2* 0.5 marks = 1 mark</i>	0.5+0.5	1	
	• Lead acid battery • Nickel iron alkaline battery • Nickel cadmium alkaline battery			
I. 7	<u>Super conductor</u> Materials whose resistance drops abruptly to zero at a temperature below its critical temperature are called <u>superconductors.</u>	1	1	
I. 8	<u>Permeability</u> The property of magnetic materials which allows the flux to pass through it is called <u>permeability.</u>	1	1	
I. 9	<u>Capacitance</u> <i>Definition- 1 mark</i>	1	1	
	Capacitance of a capacitor is defined as the amount of charge required to create a unit potential difference between its plates.			
	PART B			24
II. 1	<u>Phenomenon of electric shock</u> <i>Explanation- 3 marks</i>	3	3	
	• Like metals, human body is a conductor of electricity.			

	● When the human body comes in contact with live electricity, it sends electric current through the body. ● This flow of electric current or electric charge is experienced as electric shock. ● Effect of an electric shock may vary from minor tinkle to severe burns. ● The effect of electricity depends on four factors: 1. Magnitude of voltage 2. Resistance of skin 3. Path of electric current 4. Duration for which current flow			
II. 2	<u>Safety precautions to be followed while handling electricity</u> <i>any six points</i> <i>6 * 0.5 mark = 3 marks</i>	6 * 0.5	3	
	● Avoid contact with energized electrical circuits. ● Treat all electrical devices as if they are live or energized. ● Disconnect the power source before servicing or repairing electrical equipment. ● Use only tools and equipment with non-conducting handles when working on electrical devices. ● Never use metallic pencils or rulers, or wear rings or metal watch bands when working with electrical equipment. ● Do not wear loose clothing or ties near electrical equipment; wear smart fitting overalls if possible. ● Enclose all electric contacts and conductors so that no one can accidentally come into contact with them. ● Drain capacitors before working near them and keep the short circuit on the terminals during the work to prevent electrical shock.			
II. 3	<u>RCCB schematic diagram</u> <i>Diagram with all labelling - 3 mark</i>	3	3	

	The diagram shows a transformer with a central magnetic core. A primary coil is wound on the left limb, connected to a 'SUPPLY' with lines L and N. A 'Sensing Coil' is also wound on the left limb. A 'Test Button' is connected in series with the primary coil. A secondary coil is wound on the right limb, connected to a 'LOAD' and grounded. A 'Relay' is connected to the sensing coil and grounded. A dashed line indicates a connection from the relay to the ground of the load.			
II. 4	An electric iron takes 4 amperes at 250 volts. Find the number of units consumed if the iron is used for 90 minutes. <i>Power calculation- 1mark</i> <i>Energy calculation 2 marks</i>	1+2	3	
	$I = 4A, V = 250V, t = 90 \text{ minutes}$ Power, $P = VI = 250 \times 4 = 1000W = 1 \text{ KW}$ $T = 90 \text{ minutes} = 1.5 \text{ hours}$ Electrical energy = power x time = $1 \text{ KW} \times 1.5 \text{ h} = 1.5 \text{ KWh}$ $= 1.5 \text{ units}$			
II. 5	<u>Requirements of a good heating element</u> <i>any six points</i> <i>6 * 0.5 mark = 3 marks</i>	3	3	
	• High specific resistance • High melting temperature • Low temperature coefficient of resistance • High oxidizing temperature • Positive temperature coefficient of resistance • High ductility and flexibility • High mechanical strength			
II. 6	<u>Construction of a lead acid battery</u>	2+1	3	

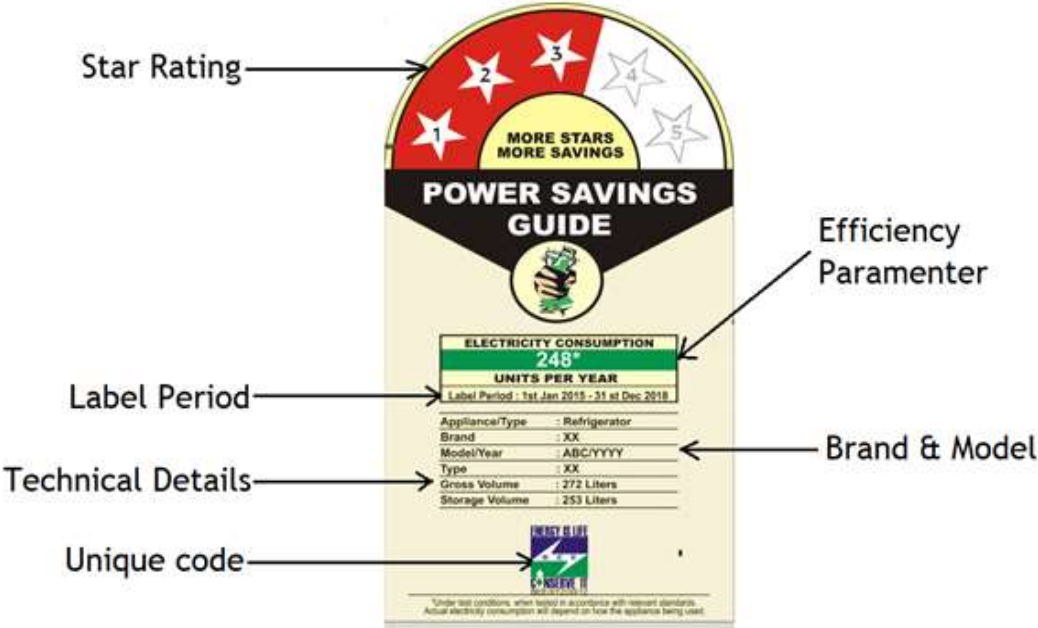
	<i>figure- 2 marks</i> <i>explanation- 1 mark</i>  • The lead acid battery consists of lead acid cells connected in series inside a container. • Each cell consists of a positive and a negative plate immersed in sulphuric acid. • The positive plate (anode) consists of lead peroxide and negative plate (cathode) consists of lead.			
II. 7	<u>Indications of a fully charged lead acid battery</u> <i>any three points</i> <i>3* 1 mark =3 marks</i> (i) Gassing When the cell is fully charged, it freely gives off hydrogen at cathode and oxygen at anode. This process is known as gassing. (ii) Voltage The voltage ceases to rise when the cell becomes fully charged. The approximate value of the emf is 2.1 V. (iii) Specific gravity of electrolyte The value of density when the cell is fully charged is 1.21 and 1.18 when discharged up to 1.8V. (iv) Colour The colour plates, on full charge is deep chocolate brown for positive plate and clear slate gray for negative plate and the cell looks quite brisk and alive.	3	3	
II. 8	<u>Factors affecting capacity of battery</u> <i>Any Six points</i> <i>6 * 0.5mark =3marks</i> • Material and structure of anode and cathode plates • Electrolytes in battery • Separator material and its properties • Distance between plates • Purity of material			

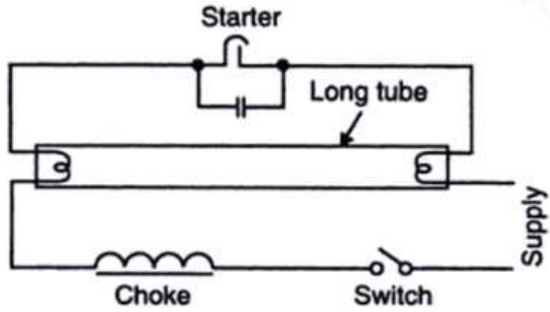
	• Charging and discharging cycle • Individual cell voltage • Recharging process • Deep discharge • Regular maintenance • Physical shocks and vibrations • Ambient temperature			
II. 9	<u>Select a suitable material for electrical wires and list any five electrical properties.</u> <i>List anyone - 0.5 mark any five properties 5 * 0.5 mark = 2.5 marks</i>	0.5+2.5	3	
	Materials: Copper, Aluminium <u>Properties of conductor</u> • Low resistivity/ High conductivity • Low temperature coefficient of resistance • Good thermal conductivity • Malleability • Should be highly ductile • Easy to get in shapes, rods and wires • Environmental stability • Compliant to manufacturing process			
II. 10	<u>Coulomb's laws of magnetic force</u> <i>Statement - 2marks Mathematical Expression- 1mark</i>	2+1	3	
	Force between two magnetic poles placed in a medium is • Directly proportional to their field strengths. • Inversely proportional to the square of the distance between them. • Inversely proportional to the absolute permeability of the surrounding medium.			
	PART C			42
III	<u>Specification table of (i) Ceiling fan, (ii) Washing machine.</u> <i>Specification ceiling fan - 4 Marks washing machine- 3 Marks</i>	4+3	7	
	(i) Ceiling fan			

Sl. No.	Specifications	Ratings
1	Rated voltage	220-240V, AC, 50 Hz
2	Rated speed	340 – 370 rpm
3	Rated air delivery	210 m ³ /min
4	Power consumption	Not more than 63 watts
5	Power factor	More than 0.9
6	Class of insulation	B
7	Capacitor	1.6 - 2.0 mF

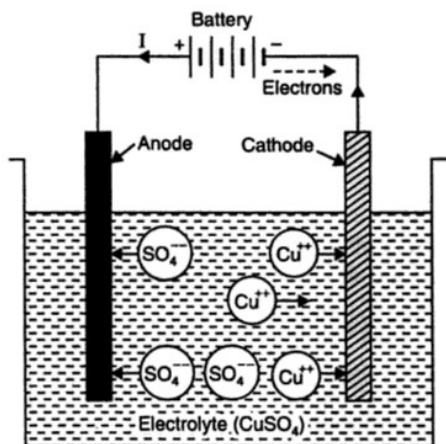
(ii) Washing machine

Sl. No.	Specifications	Ratings
1	Rated voltage	220 - 240V
2	Load capacity	8 kg
3	Spin speed	1400 rpm
4	Energy label	4 - 5
5	Energy consumption	250 KWh/year
6	Water consumption	8.5L/kg
7	Type	Front/Top load
8	Motor	DC synchronized brushless

IV	<u>BEE star labelling and various descriptions</u> <i>schematic diagram- 4 marks</i> <i>description- 3 marks</i>	4+3	7	
				
	Star Rating When comparing similar sized products look for more stars and save money. Efficiency parameter Lower electricity consumption means higher savings. Label Period Label period is the period for which standard is applicable. Standards are revised with time. Brand & Model Type of appliance, brand, model, year of manufacturing Technical details Technical parameters with respect to the appliance type			

	Unique code Unique code given by a standardizing agency.			
V	<u>Construction and working of a fluorescent lamp</u> <i>Schematic diagram - 2 marks</i> <i>construction- 2 marks</i> <i>working- 3 marks</i>	2+3+3	7	
	 Construction • The lamp is in the form of a long tube, coated inside with phosphor. • The tube contains a small amount of mercury and a small quantity of argon gas at a pressure of 2.5mm of mercury. • At each end of the tube the electrodes are spiral form made of tungsten and coated with an electron emitting material. • A choke is connected in series with the tube filament. • The filament is connected to a starter which is a small bimetallic strip. Working • When the starter is cold, the electrodes are open. When supply is given, full voltage acts on the starter. • A glow discharge is set up in the starter which warms the electrodes and causes the bimetallic strip to bend and touch the electrodes. • Circuit becomes complete and current flows causing emission of electrons. • Now the starter voltage falls to zero and the bimetallic strip cools down. • The starter switch is open and interrupts the current in the circuit causing a voltage surge of 1000V in the choke. • This voltage produces the flow of electrons between the lamp electrodes and the lamp lights up immediately.			

VI	<u>Working principle of LED lamp</u> <i>diagram- 4 marks</i> <i>explanation- 3 marks</i> The diagram illustrates the working principle of an LED lamp. It shows a P-type semiconductor (left) and an N-type semiconductor (right) connected in forward bias. The P-type region contains holes (red dots) and the N-type region contains electrons (black dots). Arrows indicate the movement of carriers towards the junction. Below the physical structure, the energy band diagram is shown. The Conduction Level is at the top, the Fermi Level is in the middle, and the Valence Band is at the bottom. The Forbidden Gap is the energy difference between the Conduction Level and the Valence Band. Electrons move from the Conduction Level to the Valence Band, and holes move from the Valence Band to the Conduction Level. This process is labeled as 'Recombination' and results in the emission of 'Light' (represented by a red wavy arrow). • The working of the LED depends on the quantum theory. • The quantum theory states that when the energy of electrons decreases from the higher level to lower level, it emits energy in the form of photons. • The energy of the photons is equal to the gap between the higher and lower level. • The LED is connected in the forward biased, which allows the current to flow in the forward direction. • The flow of current is because of the movement of electrons in the opposite direction. • The recombination shows that the electrons move from the conduction band to valence band and they emit electromagnetic energy in the form of photons. • The energy of photons is equal to the gap between the valence and the conduction band.	4+3	7	
VII	<u>Basic principle of electrolysis with a neat sketch</u> <i>diagram- 3 marks</i> <i>explanation- 4 marks</i>	3+4	7	

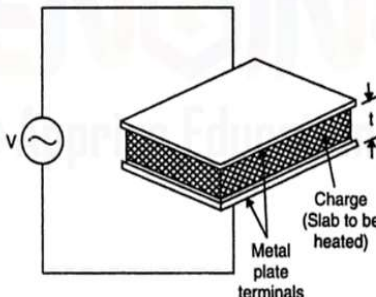
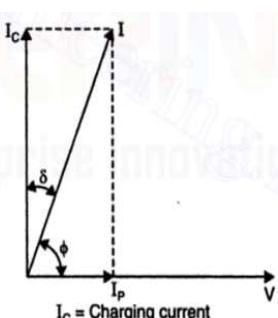


- When a compound formed by electrovalent bond is dissolved in water which has high dielectric constant, it results in the weakening of the electrostatic force of attraction between the ions.
- Consider the case of copper sulphate (CuSO_4) dissolved in water.
- It dissociates positively charged copper ions (Cu^{++}) and negatively charged sulphate ions (SO_4^{--}) moving freely in the solution.
- If two electrodes are placed and one of them is made positive and the other negative, the positively charged ions travel towards cathode and the negatively charged ions travel towards anode.
- Positively charged copper ions reaching cathode will take two electrons from it and become copper atom.
- Negatively charged sulphate ions reaching anode will give up two electrons.
- Copper is deposited at cathode and sulphate ions at anode reacts with water to form oxygen

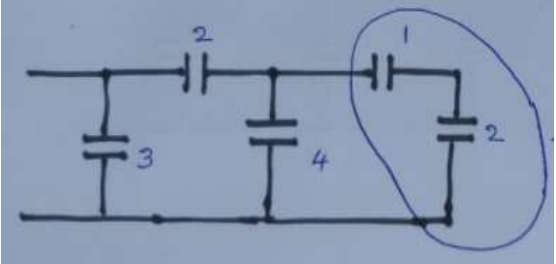
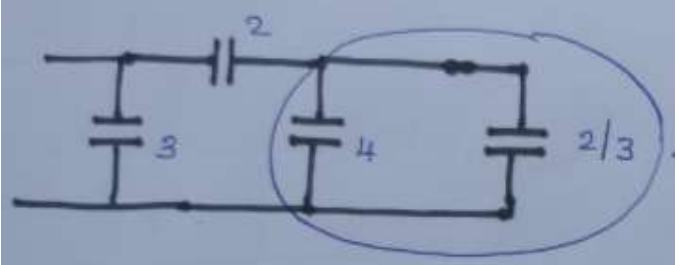
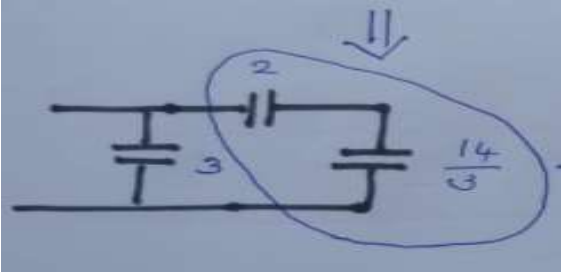
$$2\text{H}_2\text{O} + 2\text{SO}_4 = 2\text{H}_2\text{SO}_4 + \text{O}_2$$
- Oxygen is liberated as gas at anode and H_2SO_4 is formed
- If the cathode is made of Cu, the sulphuric acid attacks it forming copper sulphate and liberating hydrogen.

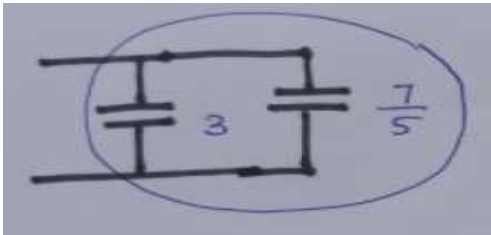
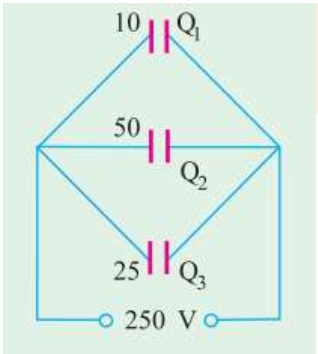
$$\text{H}_2\text{SO}_4 + \text{Cu} = \text{CuSO}_4 + \text{H}_2$$
- During the process there is no accumulation of charge at any point in the circuit and the mass of copper deposited at the cathode is equal to that removed from the anode.
- This whole process is called electrolysis.

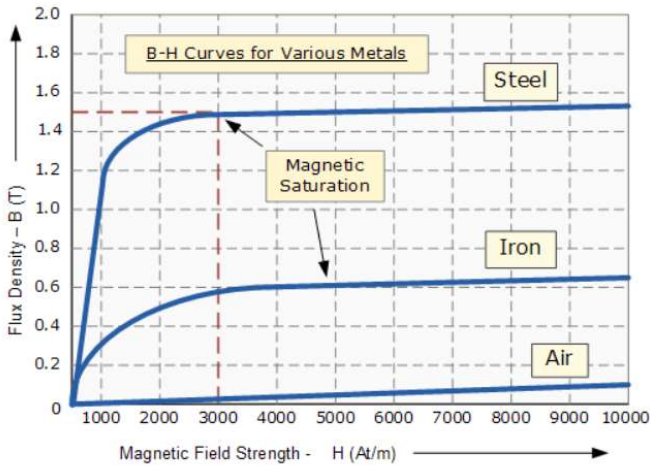
VIII	<u>Principle of dielectric heating</u>	diagram- 4 marks	4+3	7	
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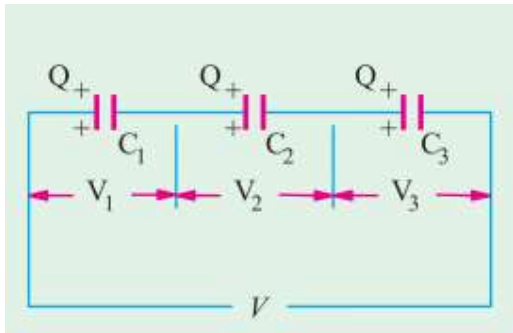
	<i>explanation- 3 marks</i>   • When a capacitor is subjected to sinusoidal voltage, the current drawn by it is never leading the voltage by exactly 90 degrees. • The angle between the current and voltage is slightly less showing there is a small in phase component of current which produces power loss in the dielectric of the capacitor • At ordinary frequency of 50Hz such loss may be small but at high frequencies the loss becomes large enough to heat the dielectric. • It is utilized in heating the dielectric. • The insulating material to be heated is placed between two conducting plates in order to form a parallel plate capacitor. • Power generated in dielectric is converted to heat.			
IX	<u>Maintenance of lead acid batteries</u> <i>any Seven points</i> <i>7 * 1mark = 7 marks</i> • Keep the top of every battery clean to prevent corrosion and leakage of current. • Keep the level of electrolyte proper in every cell of the battery. Distilled water is only added to the electrolyte till all plates are completely submerged in electrolyte. • Check the condition of electrolyte by checking specific gravity by hydrometer. • Do not charge the battery at high current otherwise due to rise in temperature excessive gassing will take place. • Do not over charge the battery. • Do not keep the battery in a discharged condition for several days. • Do not allow the short circuit of terminals of the battery. Since internal resistance is very small, heavy current flows when short circuit occurs and plates of the battery get damaged.	7	7	

X	<u>Classification of insulators</u> <i>Classification- 1 mark</i> <i>Explanation- 3*1 =3 marks</i> <i>Examples 3*1 =3 marks</i>	1+3+3	7	
	<pre> graph TD A[Physical Classification of Insulators] --> B[Solid] A --> C[Liquid] A --> D[Gaseous] </pre> Solid insulators: These insulating materials are in solid state. They are further divided into two types. They are generally used as a coating over current carrying conductor or used to make switches. (a) Hard insulators eg: Bakelite, Porcelain, Wooden plank (b) Soft insulators eg: Rubber, Polyvinyl chloride, Varnish coated papers Liquid insulators: These insulating materials are in liquid state. They are generally used in transformers, circuit breakers etc. eg: Mineral oil, Shellac, Varnish Gaseous insulators: These are the insulating material in gaseous state. Electronegative gases make good insulators since the ions rapidly combine with the ions produced in the spark. Gases with electronegative species make good insulators eg: CCl_4, CCl_2F_2, CCl_3F			
XI	Find the equivalent capacitance of the circuit shown in fig. All capacitances are in μF .	7	7	

	Step 1  1 μF and 2 μF in series $C_{eq} = (1 \times 2)/(1 + 2)$ $= \frac{2}{3} \mu$F	2		
	Step 2  4 μF and $\frac{2}{3} \mu$F in parallel $C_{eq} = 4 + (\frac{2}{3})$ $= \frac{14}{3} \mu$F	1		
	Step 3  2 μF and $\frac{14}{3} \mu$F in series $C_{eq} = (2 \times \frac{14}{3})/(2 + (\frac{14}{3}))$ $= \frac{28}{20} \mu$F $= \frac{7}{5} \mu$F	2		
	Step 4	1		

	 3μF and 7/5 μF in parallel $C_{eq} = (3 + (7/5))$ $= 22/5 \mu\text{F}$ $= 4.4 \mu\text{F}$			
	<i>Mark for unit of capacitance</i>	1		
XII	Three capacitors A, B, C have capacitances 10, 50 and 25 μF respectively. Calculate (i) charge on each when connected in parallel to a 250V supply (ii) total capacitance.  <i>(i) - 4 marks, (ii) - 3 marks</i>	4+3	7	
	(i) Each capacitor has a p.d. of 250V across it. $Q_1 = C_1V = 10 \times 250 = 2500 \mu\text{C}$ $Q_2 = C_2V = 50 \times 250 = 12500 \mu\text{C}$ $Q_3 = C_3V = 25 \times 250 = 6250 \mu\text{C}$ <i>3* 1marks each +1 mark for unit</i>	4		
	(ii) $C = C_1 + C_2 + C_3$	3		

	$= 10 + 50 + 25 = 85 \text{ uF}$ <i>2 marks for capacitance + 1 mark for unit</i>			
XIII	<u>B-H curve</u> <i>B-H curve- 4 marks</i> <i>Explanation- 3 marks</i>	3+4	7	
	 The graph illustrates the relationship between magnetic field strength (H) and flux density (B) for different materials. The Steel curve shows the highest flux density for a given field strength, reaching a saturation point around 1.5 T. The Iron curve reaches a lower saturation point of approximately 0.6 T, and the Air curve remains very low, near 0.1 T, even at high field strengths. The 'Magnetic Saturation' point is indicated where the flux density stops increasing significantly with further increases in field strength. • The magnetisation curve gives the relation between B and H for different types of core materials. • The flux density increases in proportional to the field strength until it reaches a certain value where it cannot increase any more. • This is because there is a limit to the amount of flux			

	density that can be generated by the core as all the domains in the iron are perfectly aligned. - Any further increase will have no effect on the value of flux density - The point where flux density reaches its limit is called magnetic saturation.			
XIV	Derive the expression for the equivalent capacitance when a group of capacitors are connected in series. <i>Diagram - 2 Marks</i> <i>Derivation - 5 Marks</i>	7	7	
	 C_1, C_2, C_3 = Capacitances of three capacitors V_1, V_2, V_3 = Potential difference across three capacitors. V = Applied voltage across combination C = combined or equivalent capacitance In series combination, charge on all capacitors is the same but potential difference across each is different. $V = V_1 + V_2 + V_3$			

	$\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$ $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ So for series combination of capacitors, reciprocal of equivalent capacitance is equal to the sum of reciprocals of individual capacitances.			
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Scoring Indicators

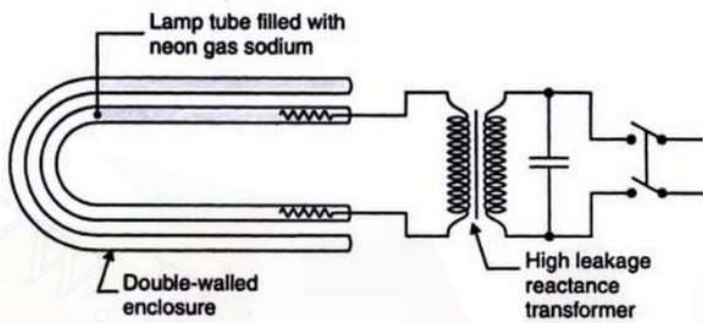
Model Question Paper II

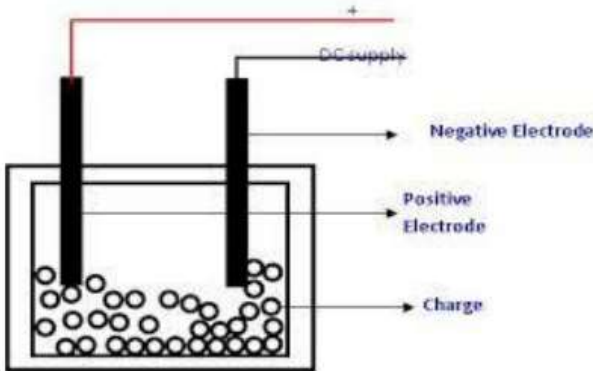
ELEMENTARY CONCEPTS OF ELECTRICAL SYSTEMS

Q No	Scoring Indicators	Split score	Sub Total	Total score
	PART A			
I. 1	<u>Effects of electric shock</u> <i>any two points 2* 0.5 mark =1 marks</i>	0.5+0.5	1	
	Burns Irregular heartbeats Seizures Tingling or prickling sensations loss of consciousness			
I. 2	<u>Nodal agency responsible to implement the star labelling program in India.</u>	1	1	
	Bureau of Energy Efficiency			
I. 3	<u>Types of a sodium vapour lamps</u> <i>List 2* 0.5 mark =1 marks</i>	1	1	
	High pressure sodium vapour lamp Low pressure sodium vapour lamp			
I. 4	<u>Convection mode of heat transfer</u> <i>Definition- 1Mark</i>	1	1	
	Convection is the transfer of heat within a fluid by mixing of one portion of the fluid with another.			
I. 5	<u>Electrodeposition</u> The process of depositing a coating of one metal over another metal or non metal electrically is called <u>electrodeposition</u> . <i>1 mark</i>	1	1	

I. 6	<u>Ampere hour efficiency</u> <i>Definition- 1Mark</i>	1	1	
	It is defined as the ratio of Ampere hour output during discharge to the Ampere hour input during charge.			
I. 7	Match the following <i>4* 0.25 mark =1 marks</i>	4*.25	1	
	(i) Copper - (d) Conductor (ii) Bakelite - (a) Insulator (iii) Cobalt - (b) Ferromagnetic material (iv) Magnesium - (c) Paramagnetic material			
I. 8	<u>Faraday's first law of electromagnetic induction</u> <i>Statement - 1 Mark</i>	1	1	
	Faraday's first law states that whenever the flux linked with a circuit changes, and emf is induced in it.			
I. 9	<u>Electric field</u> The region in which the electric line of force acts around a charge is called <i>electric field or electrostatic field.</i> <i>Anyone- 1 mark</i>	1	1	
	PART B			
II. 1	<u>Emergency procedures to be followed to rescue a person injured by an electric shock</u> <i>Any Six points</i> <i>6 * 0.5mark =3 marks</i>	3	3	
	● Don't touch the victims. ● Unplug the appliance or turn off the power to control panel ● Use a piece of wood, like a broom handle, dry rope or dry clothing, to separate the victim from the power source. ● Do not try to move a victim touching a high voltage wire.Call for emergency help. ● Keep the victim lying down. Do not move the victim if there is a suspicion of neck or spine injuries unless			

	absolutely necessary. ● If the victim has no pulse, begin cardiopulmonary resuscitation (CPR) and get medical attention.																								
II. 2	<u>Operation of fuse</u> <i>Explanation- 3 marks</i>	3	3																						
	● The action of fuse depends on the heating effect of electric current. ● In normal conditions, heat developed in the fuse element is dissipated to surrounding air. ● When fault occurs, current exceeds limiting value and excessive heat is generated. ● Fuse element heats, melts and breaks the circuit. ● Hence machines or apparatus are protected from excessive current.																								
II. 3	<u>Specification of an induction cooktop</u>	3	3																						
	<table><tr><th>Sl. No.</th><th>Specifications</th><th>Ratings</th></tr><tr><td>1</td><td>Input voltage</td><td>220 - 240V</td></tr><tr><td>2</td><td>Max Power</td><td>2000W</td></tr><tr><td>3</td><td>Min power</td><td>120W</td></tr><tr><td>4</td><td>Max temperature</td><td>2000⁰C</td></tr><tr><td>5</td><td>Min temperature</td><td>60⁰C</td></tr><tr><td>6</td><td>Time preset</td><td>1 min – 24 hours</td></tr></table>	Sl. No.	Specifications	Ratings	1	Input voltage	220 - 240V	2	Max Power	2000W	3	Min power	120W	4	Max temperature	2000 ⁰ C	5	Min temperature	60 ⁰ C	6	Time preset	1 min – 24 hours			
Sl. No.	Specifications	Ratings																							
1	Input voltage	220 - 240V																							
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4	Max temperature	2000 ⁰ C																							
5	Min temperature	60 ⁰ C																							
6	Time preset	1 min – 24 hours																							
II. 4	<u>Energy saving tips for a refrigerator</u> <i>Any Six points</i> <i>6 * 0.5mark =3 marks</i>	3	3																						

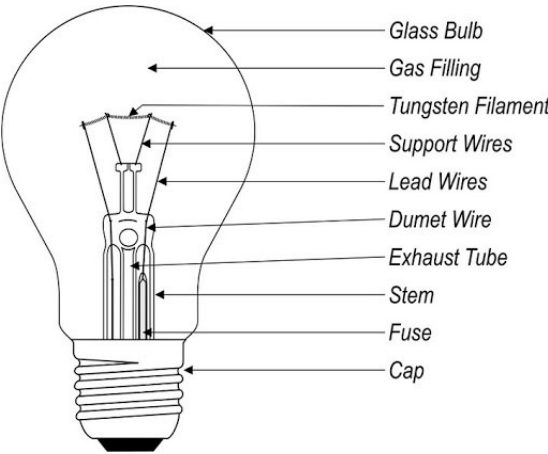
	● Allow the hot food to cool first, then put in the refrigerator. ● Do not overfill the refrigerator: Excessive loading of the refrigerator creates trouble in the circulation of cold air inside. ● Set the refrigerator temperature as per need: Lower temperature increases energy consumption. ● Don't open the refrigerator door unnecessarily: Frequent opening will increase energy consumption. ● Clean the refrigerator regularly: Cleaning the condenser coil reduces energy consumption. ● Go for star labelled refrigerators.			
II. 5	<u>Sodium vapour lamp</u> <i>Diagram-2 Marks all necessary labelling- 1marks</i>	2+1	3	
	 The diagram illustrates the components of a sodium vapour lamp. On the left, a U-shaped lamp tube is shown, labeled 'Lamp tube filled with neon gas sodium'. This tube is housed within a 'Double-walled enclosure'. The tube is connected to a 'High leakage reactance transformer', which is depicted as a rectangular box with a winding on the left and a switch on the right. The transformer is connected to a power source, represented by a battery symbol.			
II. 6	<u>Advantages of electric heating</u> <i>Any Six points 6 * 0.5mark = 3marks</i>	3	3	
	● Electric heating system is free from dirt. ● It does not produce any gas and no provision has to be made for its exit. ● Simple and accurate temperature control either manually or fully automatic. ● Electric heating is cheaper. Low initial and maintenance cost. ● Overall efficiency is very high. ● Provides better working conditions. ● No upper limit to temperature attainable.			

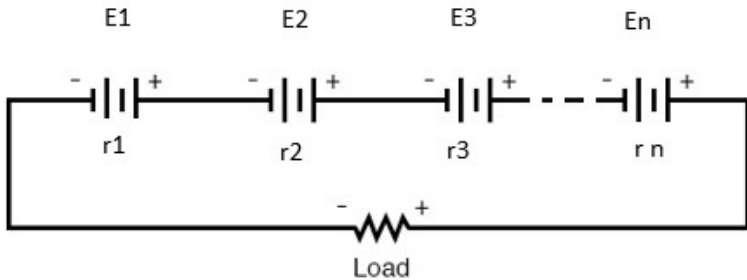
II. 7	<u>Direct resistance heating method</u> <i>figure- 2 marks</i> <i>explanation- 1 mark</i>	2+1	3	
	 The diagram shows a rectangular container filled with a granular material labeled 'Charge'. Two vertical electrodes are inserted into the charge. The left electrode is connected to a red wire labeled '+'. The right electrode is connected to a black wire labeled 'DC supply'. Labels with arrows point to the 'Negative Electrode' (the right electrode), the 'Positive Electrode' (the left electrode), and the 'Charge' (the granular material). • In this method the material or charge to be heated is treated as a resistance and current is passed through it. • The electrodes are inserted in the charge and connected to either AC or DC supply. • Heat is produced when current is passed through it.			
II. 8	<u>Factors affecting the quality of electrodeposition process</u> <i>any six points</i> <i>6 * 0.5 mark =3 marks</i>	3	3	
	• Nature of electrolyte: The electrolyte from which complex ions can be obtained provides a smooth deposit. • Current density: The deposit of metal will be uniform and fine if the current density is used at a rate higher than that at which the nuclei is formed. • Temperature: A low temperature of the solution favours formation of small crystals of metal. • Conductivity: The solution of good conductivity provides economy in power consumption. • Electrolytic concentration: By increasing the concentration of the electrolyte, higher current density can be achieved. • Addition agents: The addition of acid or other substances to the electrolyte reduces its resistance.			
II. 9	<u>Applications of lead acid battery</u> <i>Any three points</i> <i>3 * 1mark =3marks</i>	3	3	

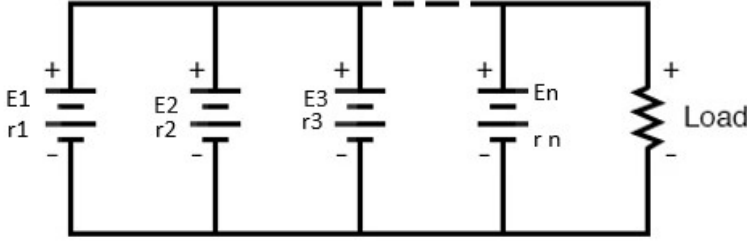
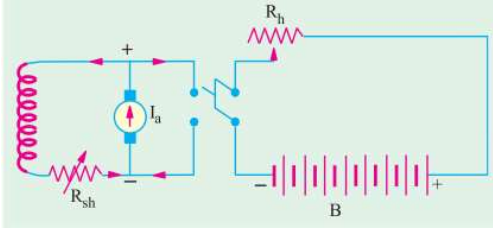
	• They are used in automobiles for lighting purposes and for starting the engines of the vehicle. • They are used for railway lighting. • Lead acid batteries are used as stand by electrical supply in operation theatres, power stations. • These batteries are used for operating protective schemes in electric power systems. • They are used in telephone exchanges, microwave stations etc.			
II. 10	<u>Coulomb's laws of electrostatics</u> <i>First law - 1 Mark</i> <i>Second law - 2 Marks</i>	1+2	3	
	First Law: Like charges of electricity repel each other whereas unlike charges attract each other. Second Law: Force exerted between two point charges I. is directly proportional to the product of their strength. II. is inversely proportional to the square of distance between them.			
	PART C			
III	Following is the electricity consumption for an office: 10 fluorescent lamps each 40 watts working 8 hours/day. 4 fans each 60 watts working 8 hours/day. 2 coolers each 100 watts working 4 hours/day. 5 lamps each 40 watts working 2 hours/day. Calculate the electricity bill for the month of April at the rate of Rs.5 per unit. <i>Energy consumption- 5 marks</i> <i>Electricity bill- 2 marks</i>	5+2	7	

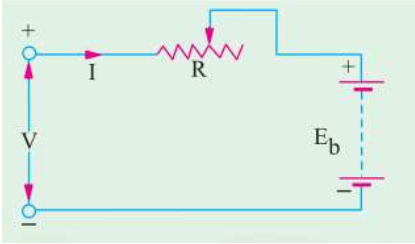
	Energy consumed by 10 tubes = $10 \times 40 \times 8 = 3200 \text{ Wh}$ Energy consumed by 4 fans = $4 \times 60 \times 8 = 1920 \text{ Wh}$ Energy consumed by 2 coolers = $2 \times 100 \times 4 = 800 \text{ Wh}$ Energy consumed by 5 lamps = $5 \times 40 \times 2 = 400 \text{ Wh}$ Total energy consumed per day = $6320 \text{ Wh} = 6.32 \text{ KWh}$	5		
	Total energy consumer for the month of April = $6.32 \times 30 = 189.6 \text{ KWh}$ Electricity bill for the month of April = $189.6 \times 5 = \text{Rs. } 948$	2		
IV	<u>Trip mechanism in MCCB</u> <i>Overload trip- 3.5 Marks</i> <i>Short circuit trip-3.5 Marks</i>	3.5+3.5	7	
	MCCB usually works for three different fault conditions: (a) Overload trip ● A flow of current that exceeds the rated current with predefined time limit such a fault is called overload. ● The bimetallic contact involves in overload operation of the breaker; under the normal condition, it allows the current flow. ● If the current flow exceeds the predefined value, then it will get bend and finally, it will engage the tripping mechanism. ● The trip mechanism opens the breaker. ● Also, bimetallic contact will not allow the breaker to reset instantly. (b) Short circuit/ instantaneous trip ● An electromagnetic coil involves short circuit/instantaneous/earth fault protection of the breaker. ● Under normal conditions, the CT generates less current			

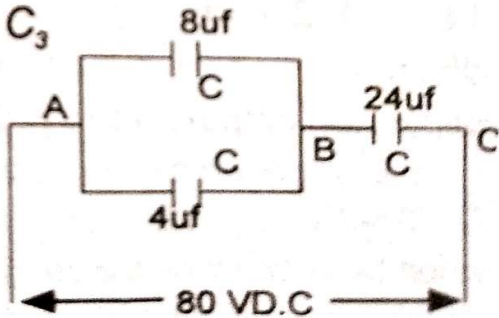
	hence the electromagnetic field is generated by the coil is not enough to pull the plunger. - During short circuit or instantaneous fault conditions, the CT generates high current and the coil creates a strong magnetic field. - Hence the coil pulls the plunger and it will trip the circuit instantly.			
V	<u>Electrolysis- Definition and Applications</u> <i>Definition- 1 mark, Any four applications- 4*1.5 = 6 marks</i>	7	7	
	The process in which electrical energy can produce chemical changes is called electrolysis. The major applications of electrolysis are: (i) Electrodeposition The process of depositing a coating of one metal over another metal or non metal electrically is called the electro deposition. (ii) Electroplating Electroplating is an art of depositing a superior or a more noble metal on an inferior or a base metal by means of electrolysis of an aqueous solution of a suitable electrolyte (iii)Electro Metallisation It is the process of depositing metal on a conducting base for decoration and for protective purposes. (iv)Electro Facing It is a process of coating of metallic surface with a harder metal by electrodeposition in order to increase its durability. (v) Electroforming Electroforming is the reproduction of objects by electrodeposition on some sort of a mould or form. (vi) Electrotyping Electrotyping is a special application of electroforming and it is used to reproduce printing, setup type, engraving and medals etc.			
VI	<u>Incandescent lamp.</u> <i>schematic diagram - 4 marks explanation- 3 marks</i>	4+3	7	

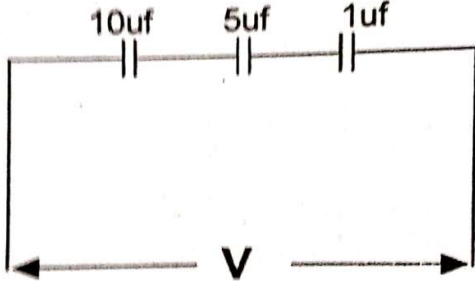
	 ● When an object is made hot, the atoms inside the object become thermally excited. ● If the object does not melt, the outer orbit electrons of the atoms jump to higher energy level due to the supplied energy. ● The electrons on these higher energy levels are not stable, they again fall back to lower energy levels. ● While falling from higher to lower energy levels, the electrons release their extra energy in a form of photons. ● These photons are then emitted from the surface of the object in the form of electromagnetic radiation. ● This radiation will have different wavelengths. ● A portion of the wavelengths is in the visible range of wavelengths, and a significant portion of wavelengths are in infrared range. ● The electromagnetic wave with wavelengths within the range of infrared is heat energy and the electromagnetic wave with wavelengths within visible range is light energy. ● In incandescent lamps, electric current flows through a thin and fine filament to produce visible light. The current rises the temperature of the filament to such extent that it becomes luminous.			
VII	<u>Classification of insulators based on temperature</u> <i>list 7 classifications-7 * 0.5 = 3.5 Marks</i> <i>Operating temperature -7 * 0.5 = 3.5 Marks</i> ● Class Y insulation: Withstands a temperature of up to 90 degree celsius; typically made of cotton, silk or paper. ● Class A insulation: Withstands a temperature of up to 105 degree celsius; reinforced Class Y materials with impregnated varnish or insulation oil. ● Class E insulation: Withstands a temperature of up to 120 degree celsius. ● Class B insulation: Withstands a temperature of up to 130 degree celsius. This has a form that inorganic material is	7	7	

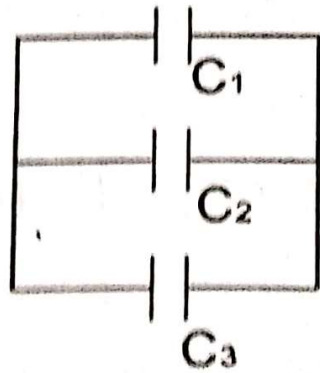
	hardened with adhesives. ● Class F insulation: Withstands a temperature of up to 155 degree celsius; made of Class B materials that are adhesives, silicon and alkyd resin varnish. ● Class H insulation: Withstands a temperature of up to 180 degree celsius; made of inorganic material glued with silicon resin ● Class C insulation: Withstands a temperature of up to 180 degree celsius or higher; made of 100% inorganic material.			
VIII	<u>Operation of storage batteries</u> <i>series connection- figure- 1.5 marks, explanation- 2 marks, parallel connection- figure- 1.5 marks, explanation- 2 marks</i>	3.5+3.5	7	
	Series Batteries  ● When in a battery, positive terminal of one cell is connected with the negative terminal of succeeding cell, then the cells are said to be series connected or simply series battery. ● Here, overall emf of the battery is algebraic sum of all individual cells connected in series. ● But overall discharged current of the battery does not exceed the discharged current of individual cells. ● If E is the overall emf of the battery combined by n number cells and E1, E2, E3, En are the emfs of individual cells. ● <i>Then, $E = E_1 + E_2 + E_3 + \dots + E_n$</i> ● Similarly, if r1, r2, r3, rn are the internal resistances of individual cells, then the internal resistance of the battery will be equal to the sum of the internal resistance of the individual cells i.e. ● <i>$r = r_1 + r_2 + r_3 + \dots + r_n$</i>			

	Parallel Batteries  - When positive terminals of all cells are connected together and similarly negative terminals of these cells are connected together in a battery, then the cells are said to be connected in parallel. - These combinations are also referred as parallel batteries. - If emf of each cell is identical, then the emf of the battery combined by n numbers of cells connected in parallel, is equal to the emf of each cell. The resultant internal resistance of the combination is, $\left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \dots + \frac{1}{r_n} \right)^{-1}$ - The current delivered by the battery is sum of currents delivered by individual cells.			
IX	<u>Charging systems of batteries</u> <i>circuit diagrams(2 methods)2 * - 2 marks</i> <i>explanation(2 methods)- 2 * 1.5 marks</i>	4+3.	7	
	(i) Constant current system  - In this method, the charging current is kept constant by varying the supply voltage to overcome the increased back emf of cells. - If charged of dc supply, the current is controlled by varying the rheostat connected in the circuit. - The value of charging current should be so chosen that there would be no excessive gassing during final stages of charging.			

	(ii) Constant voltage system  • In this method, the voltage is kept constant but it results in very large changing current in the beginning when the back emf of the cells is low and a small current when their back emf increases on being charged. • With this method, time of charging is almost reduced to half. • It increases the capacity by approximately 20% but reduces the efficiency by 10%.			
X	<u>Classification of magnetic materials</u> <i>classification- 1 mark</i> <i>Explanation- 3*2 marks = 6 marks</i>	1 + 6	7	
	Diamagnetic materials • They have weak, negative susceptibility to magnetic fields. • Diamagnetic materials are slightly repelled by a magnetic field and the material does not retain the magnetic properties when an external field is removed. • In diamagnetic materials all the electrons are paired so there is no permanent net magnetic moment per atom. • Most elements in periodic table including copper, silver, gold are diamagnetic. Paramagnetic materials • They have a small, positive susceptibility to magnetic fields. • These materials are slightly attracted by a magnetic field and the material does not retain the magnetic properties when the external field is removed. • Paramagnetic properties are due to the presence of some unpaired electrons, and from the realignment of electrons paths caused by the external magnetic field • Paramagnetic materials include magnesium, molybdenum, lithium etc. Ferromagnetic materials • They have a large, positive susceptibility to an external magnetic field. • They exhibit strong attraction to magnetic fields and are			

	able to retain their magnetic properties after the field has been removed. • Ferromagnetic materials have some unpaired electrons so their atoms have a net magnetic moment. • Iron, nickel and cobalt are examples of ferromagnetic materials.			
XI	Two capacitors 8uF and 4uF are connected in parallel and a third of 24uF is connected in series with them. The combination is connected to 80V DC. Draw the arrangement of capacitors and find equivalent capacitance and charge. <i>circuit diagram-2 mark</i> <i>Equivalent capacitance- 3 mark</i> <i>Charge -2 mark</i>	$\frac{2+3+2}{2}$	7	
	 Equivalent capacitance of parallel combination of 8uF and 4uF $= 8 + 4 = 12 \text{ uF}$ Equivalent capacitance between point A and C is given by $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{12} + \frac{1}{24} = \frac{3}{24}$ Equivalent capacitance, $C = \frac{24}{3} = 8 \text{ uF}$ Charge $Q = CV = 8 \times 10^{-6} \times 80 = 640 \text{ uC}$			

XII	A set of three capacitors having capacitances of 10, 5 and 1 microfarad respectively is available. What is the value of the greatest and the smallest capacitance that can be obtained from the set and draw the arrangements. <i>Series connection- 3.5 marks</i> (circuit diagram- 1.5, calculation- 2marks) <i>Parallel connection- 3.5 marks</i> (circuit diagram- 1.5, calculation- 2marks)	3.5+3.5	7	
	Lowest value of capacitance is obtained when all capacitors are connected in series.  $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ $\frac{1}{C_{eq}} = \frac{1}{10} + \frac{1}{5} + \frac{1}{1} = 1.3$ $C_{eq} = 1/1.3 = 0.7692\mu F$ Largest value of capacitance is obtained when all capacitors are connected in parallel			



$$C_{eq} = 10 + 5 + 1 = 16\mu\text{F}$$

XIII Comparison between electric and magnetic circuits.
Figure- 2 marks
*Any five points- 5*1 mark =5 Marks*

7

7

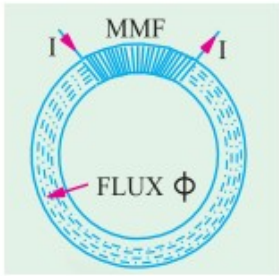
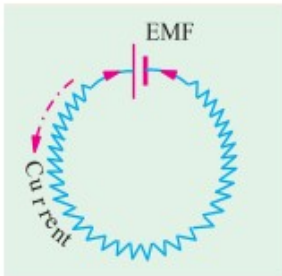
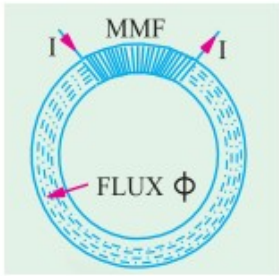
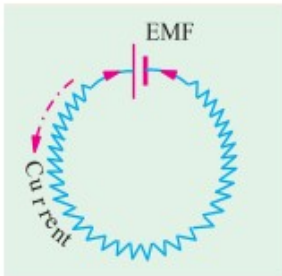
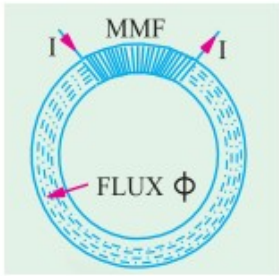
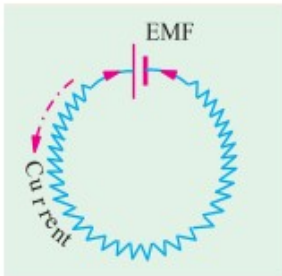
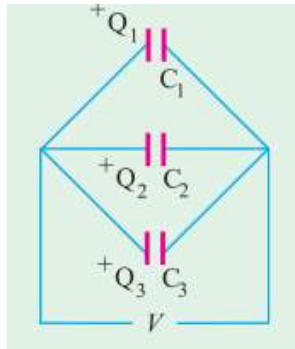
	<table> <tr> <th>Magnetic Circuit</th> <th>Electric Circuit</th> </tr> <tr> <td>  </td> <td>  </td> </tr> <tr> <td>Flux = $\frac{MMF}{Reluctance}$</td> <td>Current = $\frac{EMF}{Resistance}$</td> </tr> <tr> <td>MMF (ampere turns)</td> <td>EMF (volts)</td> </tr> <tr> <td>Flux ϕ (webers)</td> <td>Current I (amperes)</td> </tr> <tr> <td>Flux density B (Wb/m^2)</td> <td>Current density (A/m^2)</td> </tr> <tr> <td>Reluctance $S = \frac{l}{\mu A}$</td> <td>Resistance $R = \rho \frac{l}{A}$</td> </tr> <tr> <td>Permeance (= 1/re reluctance)</td> <td>Conductance (= 1/resistance)</td> </tr> <tr> <td>Reluctivity</td> <td>Resistivity</td> </tr> <tr> <td>Permeability (= 1/re reluctance)</td> <td>Conductivity (= 1/resistance)</td> </tr> </table>	Magnetic Circuit	Electric Circuit			Flux = $\frac{MMF}{Reluctance}$	Current = $\frac{EMF}{Resistance}$	MMF (ampere turns)	EMF (volts)	Flux ϕ (webers)	Current I (amperes)	Flux density B (Wb/m^2)	Current density (A/m^2)	Reluctance $S = \frac{l}{\mu A}$	Resistance $R = \rho \frac{l}{A}$	Permeance (= 1/re reluctance)	Conductance (= 1/resistance)	Reluctivity	Resistivity	Permeability (= 1/re reluctance)	Conductivity (= 1/resistance)			
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Reluctivity	Resistivity																							
Permeability (= 1/re reluctance)	Conductivity (= 1/resistance)																							
XIV	<u>Parallel connection of capacitors.</u>	2+5	7																					

Diagram - 2 Marks
Derivation - 5 Marks



C_1, C_2, C_3 = Capacitances of three capacitors

Q_1, Q_2, Q_3 = Potential difference across three capacitors.

V = Applied voltage across combination

C = combined or equivalent capacitance

In parallel combination, potential difference across each is same but charge on each is different.

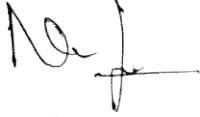
Net charge $Q = Q_1 + Q_2 + Q_3$

$CV = C_1V + C_2V + C_3V$

Hence equivalent capacitance, $C = C_1 + C_2 + C_3$

Question Papers I & II

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